QM4 Lecture Notes

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Abstract

This is my lecture note on QM4 in EPFL for 2026 spring semester.

Contents

1 Lecture 5: Gauge Invariance of QM

1.1 Classical Electromagnetism

1.1.1 Potential Convention

We here set up our convention for classical EM. The two Maxwell's equations that are source independent can be satisfied if we use a potential ansatz:

$$B(x,t) = \nabla \times A(x,t), \quad E(x,t) = -\nabla\varphi(x,t) - \partial_t A(x,t) \tag{1}$$

Note that due to the fact QM is not covariant theory, we will not use the 4-vector notation which might make life complicate. One can show that the definition is non-unique yet can be related by a gauge transformation:

$$\varphi_\chi = \varphi - \partial_t\chi, \quad A_\chi = A + \nabla\chi \tag{2}$$

1.1.2 Electrodaynamics Lagrangian

Then if we have a charged particle in the EM field, the Lagrangian is given by:

$$L(x,\dot{x},t) = \frac{1}{2}m\dot{x}^2 - e\varphi(x,t) + e\dot{x}\cdot A(x,t) \tag{3}$$

one can prove that the virational principle gives us the EoM with lorentz force. We can see that the Lagrangian is not gauge invariant:

$$L_\chi = L + e(\partial_t\chi - \dot{x}\cdot\nabla\chi) = L + e\frac{d}{dt}\chi(x(t),t) \tag{4}$$

> **Remark**
>
> Note that here is a total derivative of χ, not a partial derivative!!

Yet the action is gauge invariant up to a boundary term:

$$S_\chi = S + e\big(\chi\big(x_f, t_f\big) - \chi(x_i, t_i)\big) \tag{5}$$

1.1.3 Electrodynamics Hamiltonian

Then in order to do canonical quantization, we need to find the Hamiltonian. The canonical momentum is given by:

$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + eA(x,t) \tag{6}$$

this is different from the commonly defined "QM" momentum $\pi = m\dot{x}$. The Hamiltonian is then given by:

$$H = p\dot{x} - L = \frac{1}{2m}(p - eA(x,t))^2 + e\varphi(x,t) \tag{7}$$

We can see that the Hamiltonian is also not gauge invariant:

$$p_\chi = p + e\nabla\chi, \quad H_\chi = H - e\partial_t\chi \tag{8}$$

1.2 QM with EM Field

1.2.1 Quantization

Then we can try to define a canonical quantized theory of this. Note that the quantum theory is not unique, for the ordering of the operators shall be defined. We define the following Hamiltonian:

Definition 1.1 **QED Hamiltonian**

We use the following Hamiltonian for quantization:

$$\hat{H} = \frac{1}{2m}(\hat{p}^2 - e(\hat{p} \cdot A(x,t) + A(x,t) \cdot \hat{p}) + e^2 A(x,t)^2) + e\varphi(x,t) \tag{9}$$

note that $A(\hat{x}, t)$ is a time dependent function of the space operator now and $\hat{p}$ which is the canonical momentum operator.

We then impose the canonical commutation relation between operators:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = 0, \quad [\hat{p}_i, \hat{p}_j] = 0 \tag{10}$$

note that with this commutation relation, it is p that is the canonical momentum operator and $p = -i\hbar\nabla$ in the position basis. We can also define a Mechanical Momentum operator:

$$\hat{\pi} = \hat{p} - eA(\hat{x}, t) \tag{11}$$

This doesn't satisfy the canonical commutation relation, but it is a physical observable as we will see later.

Then we can also do path integral quantization. The transition amplitude is given by:

$$G(x_f, t_f; x_i, t_i) = \int_{x_i}^{x_f} \mathcal{D}[x] \exp\left(\frac{i}{\hbar} S[x(t)]\right) \tag{12}$$

1.2.2 Gauge Transformation of Operators

We can directly see that the Hamiltonian and the Action are not gauge invariant. Yet we want the theory to be gauge invariant. One requirement is that **the schrodinger equation should be gauge invariant**. This means that the wavefunction should also transform under gauge transformation:

$$i\hbar\partial_t \left|\psi_\chi(t)\right\rangle = \hat{H}_\chi(t) \left|\psi_\chi(t)\right\rangle \tag{13}$$

In order to have this equation to be well-defined, we must first define what O_χ means. Naively, we can just substitute the classical gauge transformation of the observable into operators. Yet this may cause problems with the Momentum Operator. if we have:

$$p_\chi \sim ???? \sim p + e\nabla\chi \tag{14}$$

then the canonical momentum operator will not commute with each other after a gauge transformation, which is not consistent with the canonical quantization result. In order to preserve the canonical commutation relation, we take the following definition for gauge transformation of momentum operator:

$$\hat{p}_\chi = \hat{p}, \quad \hat{\pi}_\chi = \hat{\pi} - e\nabla\chi(\hat{x}, t) \tag{15}$$

One should note that with this definition, the lagrangian and Hamiltonian might change differently from the classical case in the operator form.

1.2.3 Gauge Transformation of Wave Function

In order to preserve this form of schrodinger equation:

$$i\hbar\partial_t \ |\psi_\chi(t)\rangle = \hat{H}_\chi(t) \ |\psi_\chi(t)\rangle \tag{16}$$

We can have the following transformation rule of the wavefunction:

$$|\psi_\chi(t)\rangle = U_\chi(t) \ |\psi(t)\rangle \quad U_\chi(t) = \exp\left(\frac{ie}{\hbar}\chi(\hat{x}, t)\right) \tag{17}$$

If we write this gauge transformation in the position basis, we have:

$$\psi_\chi(x,t) = e^{\frac{ie}{\hbar}\chi(x,t)}\psi(x,t) \tag{18}$$

1.2.4 Gauge Transformation of Propagator

Then we can also see how the propagator transforms under gauge transformation. We know that the action transforms as Equation 5, so the propagator transforms as:

$$G_\chi(x_f, t_f; x_i, t_i) = e^{\frac{ie}{\hbar}(\chi(x_f,t_f)-\chi(x_i,t_i))}G(x_f, t_f; x_i, t_i) \tag{19}$$

This is also consistent with the fact that the wave function can be written in terms of the propagator:

$$\begin{aligned}
\psi(x_f, t_f) &= \int d^3x_i G(x_f, t_f; x_i, t_i)\psi(x_i, t_i) \\
\psi_\chi(x_f, t_f) &= \int d^3x_i G_\chi(x_f, t_f; x_i, t_i)\psi_{\chi(x_i,t_i)} \\
&= \int d^3x_i e^{\frac{ie}{\hbar}(\chi(x_f,t_f)-\chi(x_i,t_i))}G(x_f, t_f; x_i, t_i)e^{\frac{ie}{\hbar}\chi(x_i,t_i)}\psi(x_i, t_i) \\
&= e^{\frac{ie}{\hbar}\chi(x_f,t_f)}\psi(x_f, t_f)
\end{aligned} \tag{20}$$

1.2.5 U-Picture

We before make the wave function transform under gauge transformation. As Schrodinger picture to Heisenberg picture, in fact we can also define a “U-picture” where the operators transform under gauge transformation with an additional $U_\chi(t)$ operatro and the wave function is invariant.

We can define the following transformation of the operator:

$$O_\chi^{(U)}(t) = U_\chi^\dagger(t)O_\chi(t)U_\chi(t) \tag{21}$$

where $O_\chi(t)$ is naively subsituting the classical gauge transformation into the operator.

Note that consider the expectation value of an operator, if we have:

$$\hat{O}_\chi^{(U)}(t) = \hat{O}(t) \tag{22}$$

Then the observable is gauge invariant. We take all these observables to be the physical observables of the theory. In this way, we can have a gauge invariant quantum theory.

There are some examples of this transformation:

$$
\begin{aligned}
&\hat{x}_\chi = \hat{x} = \hat{x}_\chi^{(U)}, \\
&\hat{p}_\chi = \hat{p}, \quad \hat{p}_\chi^{(U)} = \hat{p} + e\nabla\chi(\hat{x}, t), \\
&\hat{\pi}_\chi = \hat{\pi} - e\nabla\chi(\hat{x}, t), \quad \hat{\pi}_\chi^{(U)} = \hat{\pi}
\end{aligned}
\tag{23}
$$

Thus we can also see here that the mechanical momentum is physical observable while the canonical momentum is not, though the canonical momentum form is gauge independent.

2 Lecture 6:

Bibliography